How much did it grow?

A bean becomes. · Curiosity day

Name _____ Date _____



Compare from the soil level. This painted stick is not a scale ruler.

PRETEND NOTEBOOK · HEIGHT ABOVE SOIL

Day	4	8	12
Height	8 cm	14 cm	23 cm

Use the pretend notebook numbers. The picture is not a ruler.

1. Day 4: 8 cm. Day 8: 15 cm. How many cm did it grow in between?

14 − 8 = _____ Show your thinking.

2. Day 12: 23 cm. From Day 8 to Day 12, how many cm did it grow?

23 + 9 = _____ Show your thinking.

3. How many days from Day 4 to Day 12?

12 − 4 = _____ Show your thinking.

4. From Day 4 (8 cm) to Day 12 (23 cm), how many cm of growth in all?

23 − 8 = _____ Show your thinking.

5. It grows 4 cm one day and 4 cm the next. How much growth together?

4 + 4 = _____ Show your thinking.

6. You watch for 7 days. You already watched 3. How many days are left?

7 − 3 = _____ Show your thinking.

WONDER DAILY

A bean becomes.

Small, but not empty

A bean rests in an open hand. "Is anything in there?" Inside, a tiny plant waits. Stored food will help it begin before leaves can cook new food.

Down and up

With water, air, and enough warmth, a root tips out first. It grows down with gravity. A shoot climbs toward light.

A new way to make food

Leaves unfold in the light. They use light, water, and carbon dioxide from the air to make food. Now the plant can make more of what it needs.

Too slow to see?

"It looks the same," says Little Wonderer. Big Wonderer opens the notebook. Yesterday's mark sits lower than today's tip. Notes catch slow change.

Look back at the beginning

They look back at the beginning. Did the bean need open leaves to start?

A question to keep

They sit by the window. Growing takes time. "What might change next?" Little Wonderer keeps a new question for tomorrow.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

Tell the bean's small story.

A bean becomes. · Curiosity day

Name _____ Date _____

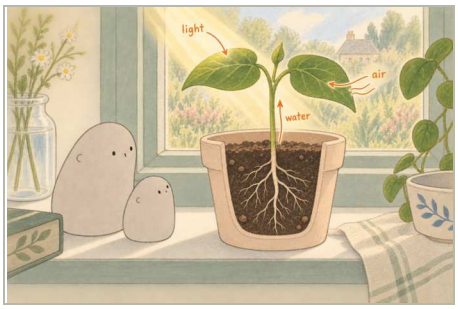
Choose a scene. Look closely, say your idea, then draw or write what the picture helped you understand.



A beginning inside



Two ways to grow



Leaves in the light

Draw the seed, root, or leaf you chose. Label one part and add an arrow to show what it does.

seed · stored food · root · shoot · leaf · light · water

Inside the bean, … What is kept there?

The root grows … Where does it go? What does it take in?

The leaf uses … What helps it make food?

Read your words aloud. Can you point to the part they explain? Add a capital letter and an ending mark if you are writing.

A bean becomes.

01 What is waking up inside this little bean?



An imagined window into the soil. The seed is enlarged to help us see.

Look closer. Look inside the seed coat. What might those two pale halves be keeping safe?

A little explanation. A tiny plant is already inside. The bean holds food to help it begin.

Try this. Curl one hand like a seed. Slowly uncurl a finger for its first root.

02 How does a root find its way when it cannot see?



Roots and shoots respond to their surroundings. Plants do not need eyes.

Look closer. Follow the white root down. Now follow the little shoot up. What do you notice?

A little explanation. The root does not need eyes. It responds to gravity, which helps it grow down.

Try this. Follow the two directions in the air: root down, shoot up.

03 What can a leaf do that a seed cannot?



Leaves make new food using light, water, and carbon dioxide from the air.

Look closer. Find the three notes around the leaves. Where does each arrow lead?

A little explanation. Green leaves use light, water, and part of the air to make new food. The seed began with food stored inside.

Try this. Follow a path from the roots to a leaf. Then point to the light.

04 How could we notice a change too slow to watch?



Compare from the soil level. This painted stick is not a scale ruler.

Look closer. Find the earlier mark and today's tip. What could a drawing help us remember?

A little explanation. We can compare notes from different days. Growth can happen too slowly for our eyes to catch it happening.

Try this. Hold a finger at the earlier mark. With your other hand, show the space up to today's tip.